

Name: _____ Partner: _____ Period: _____

Data Dispatch Agency

Your team works at the Data Dispatch Agency. Four requests just arrived, each describing a real-world situation that needs a function model. For each dispatch, your team must (1) select the best model type, and (2) state the key assumptions and domain restriction. Work through the tier that matches your group's assignment.

The four dispatches:

- **Dispatch 1.** A city tracks its population each year. The population increases by approximately 850 people per year for the past decade.
- **Dispatch 2.** A farmer wants to fence a rectangular pen using 200 m of fencing. She records the area for different choices of one side length.
- **Dispatch 3.** A warehouse stores boxes stacked in cubic arrangements. The total number of boxes for a stack of side length n units is recorded.
- **Dispatch 4.** A ride-share app charges \$2.50 per mile for the first 5 miles, then \$1.75 per mile for each mile after that.

Tier R — Remediate

For each dispatch, **circle the best model type** from the three choices given, then complete the assumption sentence.

Dispatch	Best model (circle one)			Complete the assumption
1	Linear	Quadratic	Cubic	"This model assumes the population grows by a constant amount each year."
2	Linear	Quadratic	Cubic	"This model assumes the pen is a rectangle and the perimeter is exactly 200 m."
3	Linear	Quadratic	Cubic	"This model assumes the stacks form perfect cube shapes."
4	Linear	Quadratic	Piecewise	"This model assumes the rate changes at exactly 5 miles."

Teacher Note

Dispatch 3 is cubic because n^3 boxes fit in a cube of side n . The volume/3-D clue justifies cubic.

Tier A — Apply

Dispatch 1 table:

Year (x)	Population (y)	Δy
0	12,000	—
1	12,850	850
2	13,700	850
3	14,550	850

Model type: **Linear** Degree: **1**

Assumption: **The population increases by the same amount each year (constant rate of change).**

Domain restriction: $x \geq 0$ (years since tracking began; no negative years)

Dispatch 2 table:

ℓ	A	ΔA	$\Delta^2 A$
10	900	—	—
20	1600	700	—
30	2100	500	-200
40	2400	300	-200
50	2500	100	-200

Model type: **Quadratic** Degree: **2**

Domain restriction: $0 < \ell < 100$ (length positive; other side = $100 - \ell > 0$)

Teacher Note

$A = \ell(100 - \ell) = 100\ell - \ell^2$ with perimeter 200 m means each pair of sides sums to 100 m. Second differences are all -200 (inputs spaced by 10, so $2a \cdot 10^2 = -200$ gives $a = -1$, consistent with $-\ell^2$).

Tier E — Extend

1. Dispatch 4 — Piecewise model.

Piece 1 (applies when $0 \leq m \leq 5$): $C(m) = 2.50m$

Function type for Piece 1: **linear**

Piece 2 (applies when $m > 5$): $C(m) = 2.50(5) + 1.75(m - 5) = 1.75m + 3.75$

Function type for Piece 2: **linear**

2. Two assumptions:

Assumption 1: **The per-mile rate is exactly constant within each piece (no surge pricing).** Why it matters: **If rates fluctuate, the piecewise linear model is no longer valid.**

Assumption 2: **The distance is measured in whole or fractional miles with no minimum charge beyond the stated rates.** Why it matters: **Many services charge minimums; ignoring them changes the model's starting value.**

3. Yes, the reasoning is mathematically sound. Because the number of boxes equals n^3 (a cubic formula), the third differences of the output data will be constant and nonzero, confirming a degree-3 (cubic) polynomial model.